

 INAF ISTITUTO NAZIONALE DI ASTRONOMIA OSSERVATORIO ASTRONOMICOMI DI ARCETRI	FATE MONTHLY REPORT Doc.No : FATE-001 Version : 1.0 Date : 14-03-2024	 CONSORZIO LaMMA
--	---	---

FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
February 2024



Contents

0.1 Title for Contingency tables of Night 1 2

0.2 Title for Contingency tables of Day 1 13

0.3 Title for Contingency tables of Night 2 18

0.4 Title for Contingency tables of Day 2 22

0.5 Title for Contingency tables of Night 3 24

0.6 Title for Contingency tables of Day 3 28

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	2475	1100
	$4.2 < ws < 7.6$	1491	1910
	$ws > 7.6$	169	816
Number of points 12187 ; PC=63.6%; EBD=2.2%; POD1=59.9%; POD2=49.9%; POD3=79.7%			

Table 1: night1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	3847	316
	$4.2 < ws < 7.6$	288	3287
	$ws > 7.6$	0	223
Number of points 12187 ; PC=91.7%; EBD=0.0%; POD1=93.0%; POD2=85.9%; POD3=95.7%			

Table 2: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.1 Title for Contingency tables of Night 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	10818	120
	12.0 < ws < 18.0	63	1007
	ws > 18.0	0	14
Number of points 12187 ; PC=98.1%; EBD=0.0%; POD1=99.4%; POD2=88.3%; POD3=81.8%			

Table 3: night1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	4922	317	16		70
	East	420	595	282		5
	South	62	216	2074		200
	West	176	58	82		157

Number of points 9652 ; PC=80.3%; EBD=1.5%; POD1=88.2%; POD2=50.2%; POD3=84.5%; POD4=36.3%

Table 4: night1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	5491	110	0		10
	East	79	1025	28		0
	South	0	51	2420		21
	West	10	0	6		401

Number of points 9652 ; PC=96.7%; EBD=0.0%; POD1=98.4%; POD2=86.4%; POD3=98.6%; POD4=92.8%

Table 5: night1 - Contingency table for wind direction (degree) processed with AR (1H)

FATE - Monthly Report

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1885	593
	7.1 < rh < 15.3	1680	3100
	rh > 15.3	232	1143
Number of points 12072 ; PC=66.0%; EBD=2.2%; POD1=49.6%; POD2=64.1%; POD3=86.9%			

Table 6: night1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	3489	170
	7.1 < rh < 15.3	307	4542
	rh > 15.3	1	124
Number of points 12072 ; PC=93.9%; EBD=0.0%; POD1=91.9%; POD2=93.9%; POD3=96.2%			

Table 7: night1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	11677	34
	50.0 < rh < 70.0	27	297
	rh > 70.0	0	7
Number of points 12072 ; PC=99.4%; EBD=0.0%; POD1=99.8%; POD2=87.9%; POD3=100.0%			

Table 8: night1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2608	508	0
	1.9 < pwv < 3.5	507	2359	669
	pwv > 3.5	8	470	4563
Number of points 11692 ; PC=81.5%; EBD=0.1%; POD1=83.5%; POD2=70.7%; POD3=87.2%				

Table 9: night1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	3050	81	0
	1.9 < pwv < 3.5	73	3213	57
	pwv > 3.5	0	43	5175
Number of points 11692 ; PC=97.8%; EBD=0.0%; POD1=97.7%; POD2=96.3%; POD3=98.9%				

Table 10: night1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1652	1075	271
	0.6 < see < 0.8	533	850	811
	see > 0.8	118	405	833
Number of points 6548 ; PC=50.9%; EBD=5.9%; POD1=71.7%; POD2=36.5%; POD3=43.5%				

Table 11: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2308	386	16
	0.6 < see < 0.8	280	1665	173
	see > 0.8	9	254	1849
Number of points 6548 ; PC=83.9%; EBD=0.4%; POD1=88.9%; POD2=72.2%; POD3=90.7%				

Table 12: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2206	0	192
	0.6 < see < 0.8	0	2241	192
	see > 0.8	97	89	1723
Number of points 6548 ; PC=94.2%; EBD=4.4%; POD1=95.8%; POD2=96.2%; POD3=81.8%				

Table 13: night1 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2591	3	13
	0.6 < see < 0.8	0	2290	12
	see > 0.8	6	12	2025
Number of points 6548 ; PC=99.5%; EBD=0.3%; POD1=99.8%; POD2=99.3%; POD3=98.8%				

Table 14: night1 - Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1009	301	8
	$3.4 < \tau < 5.8$	396	886	411
	$\tau > 5.8$	59	627	2129
Number of points 5826 ; PC=69.1%; EBD=1.2%; POD1=68.9%; POD2=48.8%; POD3=83.6%				

Table 15: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1379	111	0
	$3.4 < \tau < 5.8$	85	1535	149
	$\tau > 5.8$	0	168	2399
Number of points 5826 ; PC=91.2%; EBD=0.0%; POD1=94.2%; POD2=84.6%; POD3=94.2%				

Table 16: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 0.0

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1300	20	137
	$3.4 < \tau < 5.8$	105	1468	137
	$\tau > 5.8$	59	326	2411
Number of points 5826 ; PC=88.9%; EBD=3.4%; POD1=88.8%; POD2=80.9%; POD3=89.8%				

Table 17: night1 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

wavefront coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1462	1	6
	$3.4 < \tau < 5.8$	2	1785	6
	$\tau > 5.8$	0	28	2542
Number of points 5826 ; PC=99.4%; EBD=0.1%; POD1=99.9%; POD2=98.4%; POD3=99.5%				

Table 18: night1 - Contingency table for wavefront coherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1180	1141
	0.6 < glf < 0.7	544	820
	glf > 0.7	176	167
Number of points 5916 ; PC=38.3%; EBD=17.4%; POD1=62.1%; POD2=38.5%; POD3=14.0%			

Table 19: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1671	331
	0.6 < glf < 0.7	221	1508
	glf > 0.7	8	289
Number of points 5916 ; PC=79.6%; EBD=0.5%; POD1=87.9%; POD2=70.9%; POD3=81.1%			

Table 20: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1702	334
	0.6 < glf < 0.7	22	1780
	glf > 0.7	176	14
Number of points 5916 ; PC=75.4%; EBD=18.3%; POD1=89.6%; POD2=83.6%; POD3=38.1%			

Table 21: night1 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1888	16
	0.6 < glf < 0.7	4	2101
	glf > 0.7	8	11
Number of points 5916 ; PC=98.9%; EBD=0.5%; POD1=99.4%; POD2=98.7%; POD3=97.5%			

Table 22: night1 - Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	1021	302
	$4.2 < ws < 7.6$	1485	3206
	$ws > 7.6$	93	1385
Number of points 11711 ; PC=65.4%; EBD=1.2%; POD1=39.3%; POD2=65.5%; POD3=81.3%			

Table 23: day1 - Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
Model data	$ws < 4.2$	2283	231
	$4.2 < ws < 7.6$	316	4396
	$ws > 7.6$	0	266
Number of points 11711 ; PC=90.6%; EBD=0.0%; POD1=87.8%; POD2=89.8%; POD3=93.2%			

Table 24: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.2 Title for Contingency tables of Day 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
Model data	ws < 12.0	10057	116
	12.0 < ws < 18.0	97	1043
	ws > 18.0	0	30
Number of points 11711 ; PC=97.5%; EBD=0.0%; POD1=99.0%; POD2=87.7%; POD3=87.2%			

Table 25: day1 - Contingency table for wind speed (ms^{-1}) processed with AR (1H) and thresholds (12.0,18.0)
 ms^{-1}

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	3523	255	1		80
	East	71	176	49		0
	South	12	33	1329		266
	West	1685	10	144		2774

Number of points 10408 ; PC=75.0%; EBD=0.2%; POD1=66.6%; POD2=37.1%; POD3=87.3%; POD4=88.9%

Table 26: day1 - Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	5034	49	0		155
	East	41	410	4		0
	South	0	15	1479		32
	West	216	0	40		2933

Number of points 10408 ; PC=94.7%; EBD=0.0%; POD1=95.1%; POD2=86.5%; POD3=97.1%; POD4=94.0%

Table 27: day1 - Contingency table for wind direction (degree) processed with AR (1H)

FATE - Monthly Report

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	643	328
	7.1 < rh < 15.3	1186	3865
	rh > 15.3	97	1287
Number of points 11686 ; PC=71.4%; EBD=0.8%; POD1=33.4%; POD2=70.5%; POD3=89.6%			

Table 28: day1 - Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1705	142
	7.1 < rh < 15.3	221	5161
	rh > 15.3	0	177
Number of points 11686 ; PC=94.3%; EBD=0.0%; POD1=88.5%; POD2=94.2%; POD3=97.0%			

Table 29: day1 - Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	11317	17
	50.0 < rh < 70.0	18	242
	rh > 70.0	0	3
Number of points 11686 ; PC=99.6%; EBD=0.0%; POD1=99.8%; POD2=92.4%; POD3=95.5%			

Table 30: day1 - Contingency table for relative humidity (percent) processed with AR (1H) and thresholds (50,70) percent

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1512	324	3
	1.9 < pwv < 3.5	686	2604	621
	pwv > 3.5	0	416	5206
Number of points 11372 ; PC=82.0%; EBD=0.0%; POD1=68.8%; POD2=77.9%; POD3=89.3%				

Table 31: day1 - Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2111	69	0
	1.9 < pwv < 3.5	87	3236	76
	pwv > 3.5	0	39	5754
Number of points 11372 ; PC=97.6%; EBD=0.0%; POD1=96.0%; POD2=96.8%; POD3=98.7%				

Table 32: day1 - Contingency table for precipitable water vapor (mm) processed with AR (1H)

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2175	1028	89
	4.2 < ws < 7.6	1555	1768	699
	ws > 7.6	188	877	3145
Number of points 11524 ; PC=61.5%; EBD=2.4%; POD1=55.5%; POD2=48.1%; POD3=80.0%				

Table 33: night2 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.3 Title for Contingency tables of Night 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4640	345	30
	East	459	501	415
	South	40	203	1852
	West	88	77	78

Number of points 9115 ; PC=78.0%; EBD=1.7%; POD1=88.8%; POD2=44.5%; POD3=78.0%; POD4=30.5%

Table 34: night2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1568	488
	7.1 < rh < 15.3	1674	2690
	rh > 15.3	362	1292

Number of points 11402 ; PC=62.0%; EBD=3.7%; POD1=43.5%; POD2=60.2%; POD3=84.5%

Table 35: night2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2414	483
	1.9 < pwv < 3.5	509	1909
	pwv > 3.5	10	589

Number of points 11035 ; PC=80.1%; EBD=0.4%; POD1=82.3%; POD2=64.0%; POD3=88.2%

Table 36: night2 - Contingency table for precipitable water vapor (mm) in standard configuration

FATE - Monthly Report

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1598	963	340
	0.6 < see < 0.8	703	1014	816
	see > 0.8	187	601	1047
Number of points		7269 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 37: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2352	0	217
	0.6 < see < 0.8	0	2423	217
	see > 0.8	136	155	1986
Number of points		7269 ; PC=%; EBD=%; POD1=%; POD2=%; POD3=%		

Table 38: night2 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time ms		Observations		
		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1037	279	24
	3.4 < tau < 5.8	433	1035	403
	tau > 5.8	66	764	2565
Number of points		6606 ; PC=70.2%; EBD=1.4%; POD1=67.5%; POD2=49.8%; POD3=85.7%		

Table 39: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time ms		Observations		
		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1361	39	182
	3.4 < tau < 5.8	109	1555	182
	tau > 5.8	66	484	2810
Number of points		6606 ; PC=86.7%; EBD=3.8%; POD1=88.6%; POD2=74.8%; POD3=88.5%		

Table 40: night2 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1218	1271
	0.6 < glf < 0.7	666	943
	glf > 0.7	152	220
Number of points 6732 ; PC=37.6%; EBD=16.6%; POD1=59.8%; POD2=38.7%; POD3=16.4%			

Table 41: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1862	338
	0.6 < glf < 0.7	22	2077
	glf > 0.7	152	19
Number of points 6732 ; PC=76.2%; EBD=18.1%; POD1=91.5%; POD2=85.3%; POD3=38.5%			

Table 42: night2 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1051	335	61
	4.2 < ws < 7.6	1461	3042	828
	ws > 7.6	88	1391	3320
Number of points 11577 ; PC=64.0%; EBD=1.3%; POD1=40.4%; POD2=63.8%; POD3=78.9%				

Table 43: day2 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.4 Title for Contingency tables of Day 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	3253	245	7
	East	110	111	49
	South	16	30	1184
	West	1797	35	307

Number of points 10253 ; PC=71.1%; EBD=0.6%; POD1=62.8%; POD2=26.4%; POD3=76.5%; POD4=88.1%

Table 44: day2 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	504	261
	7.1 < rh < 15.3	1299	3291
	rh > 15.3	160	1840

Number of points 11552 ; PC=65.4%; EBD=1.4%; POD1=25.7%; POD2=61.0%; POD3=89.6%

Table 45: day2 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1616	361
	1.9 < pwv < 3.5	698	2297
	pwv > 3.5	7	589

Number of points 11238 ; PC=78.8%; EBD=0.2%; POD1=69.6%; POD2=70.7%; POD3=87.1%

Table 46: day2 - Contingency table for precipitable water vapor (mm) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	2166	1098	138
	4.2 < ws < 7.6	1543	1769	798
	ws > 7.6	235	789	3055
Number of points 11591 ; PC=60.3%; EBD=3.2%; POD1=54.9%; POD2=48.4%; POD3=76.5%				

Table 47: night3 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.5 Title for Contingency tables of Night 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	4628	378	41
	East	449	518	426
	South	57	182	1852
	West	93	48	94

Number of points 9153 ; PC=77.6%; EBD=1.6%; POD1=88.5%; POD2=46.0%; POD3=76.8%; POD4=27.9%

Table 48: night3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1422	606
	7.1 < rh < 15.3	1809	2678
	rh > 15.3	373	1253

Number of points 11469 ; PC=58.6%; EBD=3.5%; POD1=39.5%; POD2=59.0%; POD3=78.8%

Table 49: night3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	2520	722
	1.9 < pwv < 3.5	441	1533
	pwv > 3.5	41	724

Number of points 11102 ; PC=74.9%; EBD=1.1%; POD1=83.9%; POD2=51.5%; POD3=83.3%

Table 50: night3 - Contingency table for precipitable water vapor (mm) in standard configuration

FATE - Monthly Report

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	1614	1065	352
	0.6 < see < 0.8	735	918	881
	see > 0.8	142	592	989
Number of points		7288	PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 51: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	2371	0	236
	0.6 < see < 0.8	0	2475	236
	see > 0.8	120	100	1986
Number of points		7288	PC=%; EBD=%; POD1=%; POD2=%; POD3=%	

Table 52: night3 - Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

wavefront coherence time ms		Observations		
		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1080	290	16
	3.4 < tau < 5.8	454	944	437
	tau > 5.8	54	841	2543
Number of points		6659	PC=68.6%; EBD=1.1%; POD1=68.0%; POD2=45.5%; POD3=84.9%	

Table 53: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 0.0

wavefront coherence time ms		Observations		
		tau < 3.4	3.4 < tau < 5.8	tau > 5.8
Model data	tau < 3.4	1411	35	189
	3.4 < tau < 5.8	123	1556	189
	tau > 5.8	54	484	2807
Number of points		6659	PC=86.7%; EBD=3.6%; POD1=88.9%; POD2=75.0%; POD3=88.1%	

Table 54: night3 - Contingency table for wavefront coherence time (ms) in standard configuration and accuracy 1.22

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1153	1221
	0.6 < glf < 0.7	754	1027
	glf > 0.7	149	208
Number of points 6785 ; PC=36.4%; EBD=16.9%; POD1=56.1%; POD2=41.8%; POD3=12.8%			

Table 55: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless	Observations		
	glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	1879	265
	0.6 < glf < 0.7	28	2175
	glf > 0.7	149	16
Number of points 6785 ; PC=77.5%; EBD=18.0%; POD1=91.4%; POD2=88.6%; POD3=38.1%			

Table 56: night3 - Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

wind speed ms^{-1}	Observations		
	$ws < 4.2$	$4.2 < ws < 7.6$	$ws > 7.6$
	$ws < 4.2$	1041	371
Model data	$4.2 < ws < 7.6$	1404	3094
	$ws > 7.6$	127	1311
Number of points 11577 ; PC=63.8%; EBD=1.6%; POD1=40.5%; POD2=64.8%; POD3=77.0%			

Table 57: day3 - Contingency table for wind speed (ms^{-1}) in standard configuration

0.6 Title for Contingency tables of Day 3

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

FATE - Monthly Report

wind direction degree	Observations			
	North	East	South	West
Model data	North	3225	261	31
	East	93	100	44
	South	13	30	1191
	West	1906	36	281

Number of points 10275 ; PC=69.3%; EBD=0.8%; POD1=61.6%; POD2=23.4%; POD3=77.0%; POD4=85.1%

Table 58: day3 - Contingency table for wind direction (degree) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	357	362
	7.1 < rh < 15.3	1431	3215
	rh > 15.3	158	1832

Number of points 11552 ; PC=62.8%; EBD=1.5%; POD1=18.3%; POD2=59.4%; POD3=87.8%

Table 59: day3 - Contingency table for relative humidity (percent) in standard configuration

precipitable water vapor mm	Observations		
	pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1578	436
	1.9 < pwv < 3.5	754	2273
	pwv > 3.5	29	478

Number of points 11285 ; PC=76.7%; EBD=0.9%; POD1=66.8%; POD2=71.3%; POD3=83.8%

Table 60: day3 - Contingency table for precipitable water vapor (mm) in standard configuration